

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

--

CT
GROUP

1	5	S	
---	---	---	--

INDEX
NUMBER

--	--	--	--

BIOLOGY

9648/01

22 September 2016

Paper 1 Multiple Choice

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, CT class and index number on the Answer Sheet.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

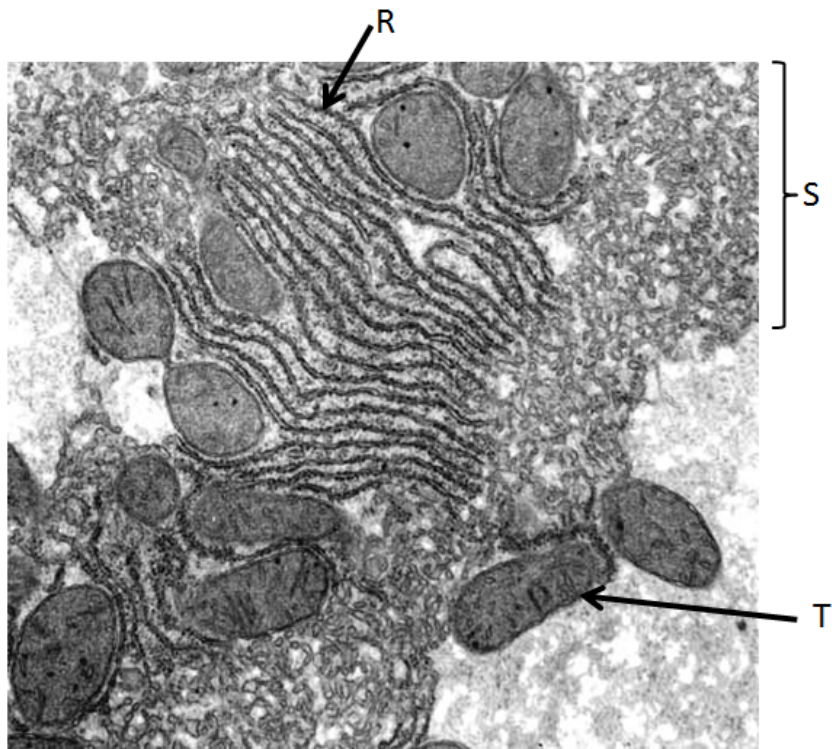
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **27** printed pages and **1** blank page.

[Turn Over

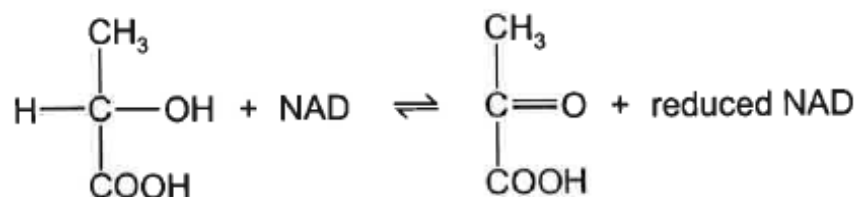
- 1 The figure below shows an electron micrograph of an eukaryotic cell.



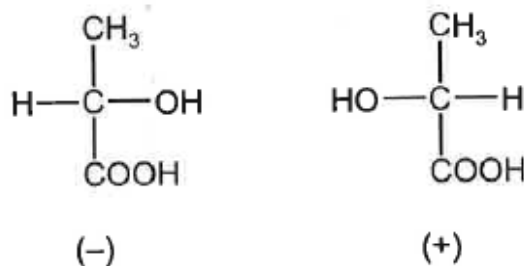
Which of the following option correctly matches the structures **R**, **S** and **T** to their respective functions?

	R	S	T
A	Involved in proteins glycosylation	Site of lipid synthesis	To convert light energy to chemical energy
B	Site of protein synthesis	Site of detoxification reaction	Supplying cellular energy
C	Site of detoxification reaction	Involved in protein glycosylation	Remove worn out organelles
D	Site of protein synthesis	Contains proteins to be secreted	Supplying cellular energy

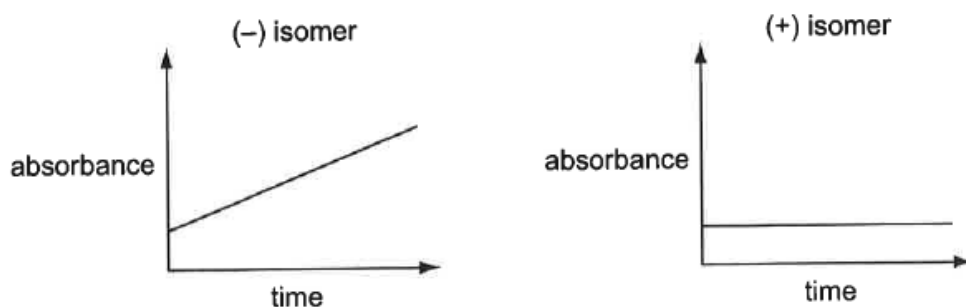
- 2 Lactic dehydrogenase catalyses the conversion of lactic acid as shown in the following equation.



Two forms (isomers) of lactic acid exist, (-) and (+), as shown below.



Reduced NAD absorbs ultraviolet light. NAD does not. The activity of bacterial lactic dehydrogenase on two different isomers of lactic acid was compared. The absorbance of ultraviolet light was measured using an ultraviolet spectrophotometer. The graphs show the results.

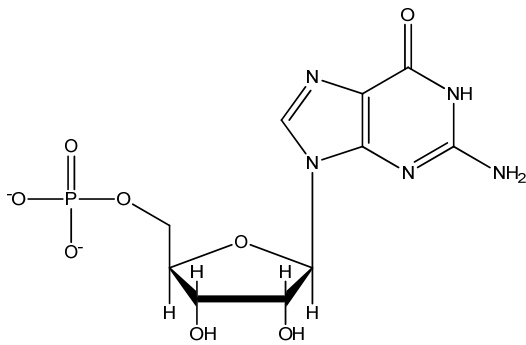
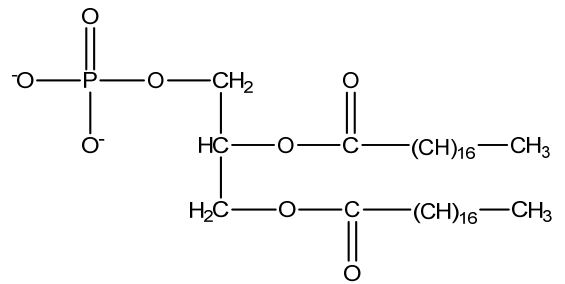
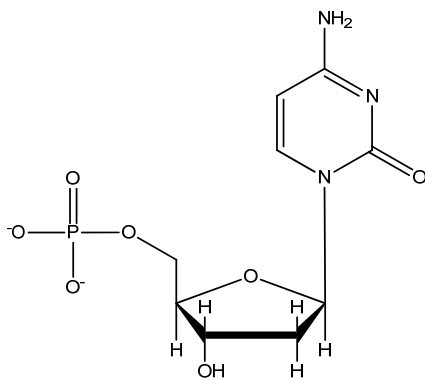
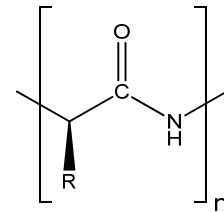


What can be concluded about bacterial lactic dehydrogenase?

- A Molecules of both isomers fit the active site.
- B Molecules of neither isomer fit the active site.
- C The enzyme is specific to the (-) isomer.
- D The enzyme is specific to the (+) isomer.

[Turn Over

- 3 A student uses centrifugation to separate the various subcellular structures of human epithelial cells by size and density.

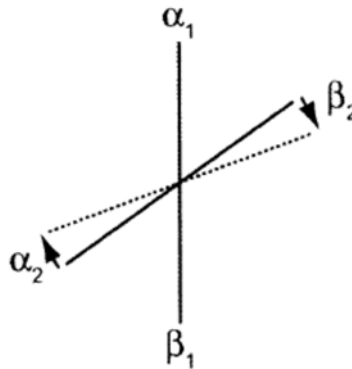
**I****II****III****IV**

Which of the following molecule(s) would you expect to find in the pellet containing the cell membrane?

- A II only
- B III only
- C II and IV only
- D I, III and IV only

- 4 A molecule of haemoglobin is made up of two α polypeptide subunits (α_1 and α_2) and two β polypeptide subunits (β_1 and β_2).

The relative positions of these subunits change when a deoxygenated haemoglobin molecule takes up oxygen. The axis joining the α_2 and β_2 subunits rotates by about 15° in relation to the axis joining the α_1 and β_1 subunits, as shown in the diagram.



Which statements about this rotation are correct?

- 1 The points of contact between the four subunits are altered.
- 2 The rotation resulting from adding one oxygen molecule to one of the subunits makes it easier to add oxygen to the other subunits.
- 3 The rotation makes different amino acids available for binding oxygen.
- 4 The rotation alters the quaternary structure of the molecule.

- A** 1, 2 and 3 only
B 1, 2 and 4 only
C 1, 3 and 4 only
D 2, 3 and 4 only

- 5 A group of diploid cells with x amount of DNA and 36 chromosomes each is capable of undergoing mitosis and meiosis.

During which stage(s) could a cell with $2x$ amount of DNA and 72 chromosomes be found?

- A** anaphase of mitosis only
B anaphase of meiosis I only
C anaphase of mitosis and anaphase of meiosis I
D anaphase of mitosis and anaphase of meiosis II

[Turn Over]

6 Below are statements that describe the control of transcription for genes encoding for enzymes in a metabolic pathway.

- 1 The genes are transcribed as a single transcription unit, with each gene having its own promoter.
- 2 The genes respond similarly to the same set of general transcription factors but respond differently to a certain set of specific transcription factors.
- 3 The genes have various combinations of control elements that enable different activators and repressors to bind and affect the rate of transcription.
- 4 The genes are found close to each other on the chromosome with chromatin remodelling as a form of transcriptional control.

Which combination of statements is true?

- A** 1 only
- B** 2 only
- C** 2 and 3
- D** 3 and 4

7 The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenized cell contents through a fractionating column. The column has short length of uracil nucleotides attached to a solid supporting material. Most molecules of mRNA that pass through the column quickly break up into small pieces and cannot be translated.

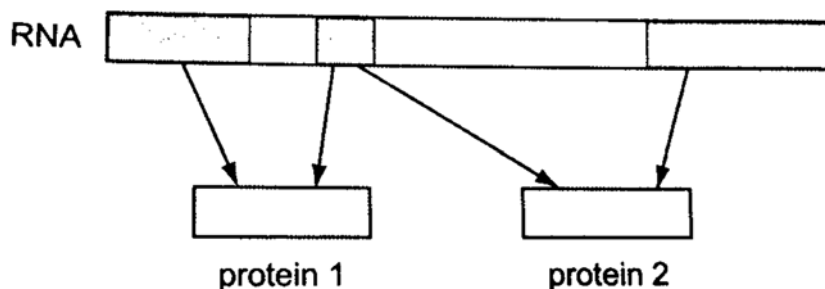
The active mRNAs that attached to the column can be separated again by appropriate treatment.

Which statements correctly describe active mRNA?

- 1 Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.
- 2 Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.
- 3 Only mRNAs with polyadenine tailing can be translated.
- 4 Polyadenine tailing destabilizes mRNA and prevents it from being broken up.

- A** 1 and 2 only
- B** 1, 2 and 3 only
- C** 3 and 4 only
- D** 1, 2, 3 and 4

- 8 RNA transcribed from a length of DNA of a chromosome was found to code for two different protein, as shown in the diagram.



Which is correct?

- A** The DNA from which this RNA was transcribed was part of a eukaryotic chromosome because this is a way of saving space in a small genome.
- B** The DNA from which this RNA was transcribed was part of a eukaryotic chromosome because introns have been edited out of the RNA.
- C** The DNA from which this RNA was transcribed was part of a prokaryotic chromosome because introns have been edited out of the RNA.
- D** The DNA from which this RNA was transcribed was part of a prokaryotic chromosome because this a way of saving space in a small chromosome.
- 9 Below are the descriptions of different gene mutations.
- 1 deletion toward the end of the code sequence
 - 2 insertion in the middle of the code sequence
 - 3 substitution close to the beginning of code sequence

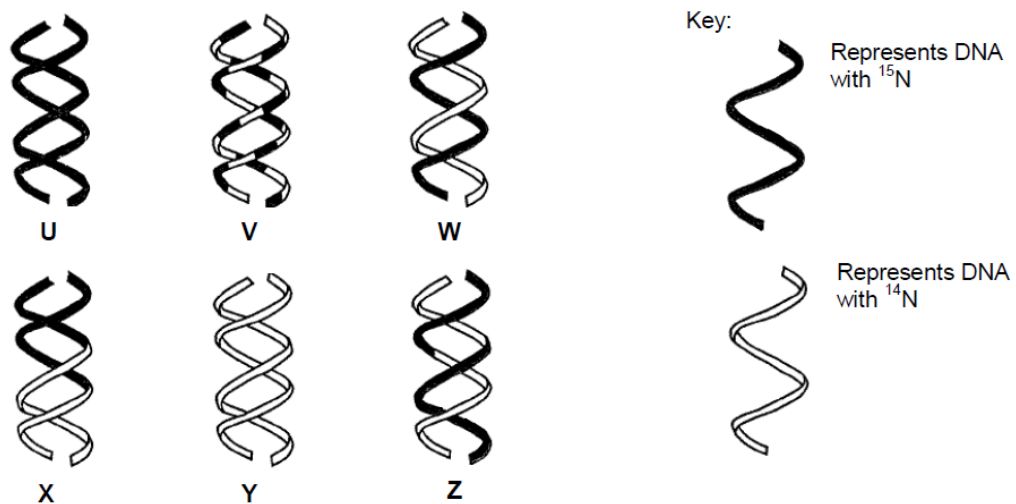
Which row correctly identifies the possible effects of these mutations on the synthesis of polypeptides?

	Premature ending of a polypeptide	A non-functional polypeptide	A polypeptide with unchanged function	A polypeptide with a different function
A	1,2,3	1,2,3	1,2,3	1,2,3
B	1,2,3	2 only	1,3 only	1,2 only
C	1,3 only	1,2,3	3 only	1,2,3
D	2,3 only	2,3 only	1,2,3	2,3 only

[Turn Over

- 10 The coding region of a gene is 135 nucleotides long, including both the start and stop codons. Which of the following would be the most likely effect of a single nucleotide deletion at position 102 in the coding region?
- A Only the active site would be affected.
 - B The entire amino acid sequence of the polypeptide would change.
 - C There would be changes in only the first 34 amino acids.
 - D There would be changes in only the last 10 amino acids.
- 11 Three experiments were carried out to investigate the mode of DNA replication in bacteria.
- Experiment 1: Bacteria were grown for many generations with only the light isotope of nitrogen, ^{14}N , and then allowed to replicate once with the heavy isotope, ^{15}N .
 - Experiment 2: Bacteria were grown for many generations with only the heavy isotope of nitrogen, ^{15}N , and then allowed to replicate once with the light isotope, ^{14}N .
 - Experiment 3: Bacteria were grown for many generations with only the heavy isotope of nitrogen, ^{15}N , and then allowed to replicate twice with the light isotope, ^{14}N .

The figure shows possible DNA molecules U to Z and indicates the varying proportion of nitrogen isotopes present in each DNA molecule.

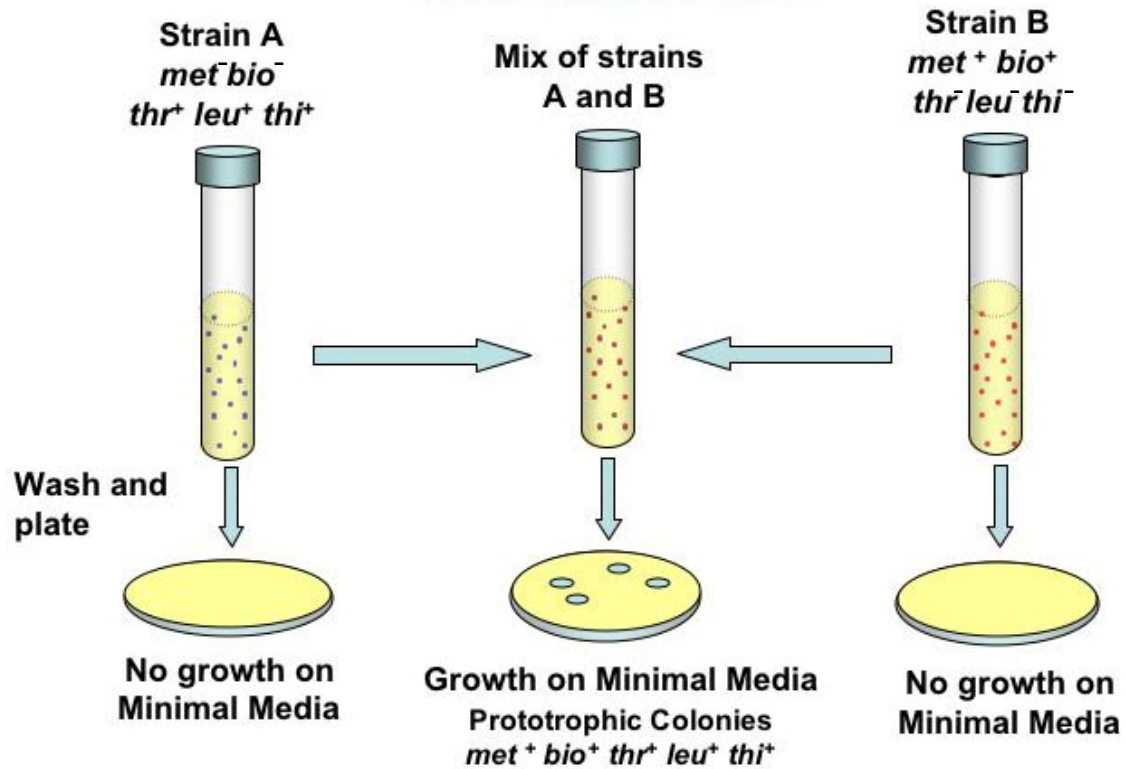


Which of the following products shows the semi-conservative mode of DNA replication?

	Experiment 1	Experiment 2	Experiment 3
A	W	W	W and Y
B	U	W	W and Y
C	W	W	X and Z
D	U	W	V and Z

12 The diagram shows an investigation into bacterial genetics.

The researchers used 2 strains of bacteria. Strain A coded for 3 essential amino acids ($\text{thr}^+ \text{leu}^+ \text{thi}^+$) while Strain B coded for 2 essential amino acids ($\text{met}^+ \text{bio}^+$). For the 2 strains of bacteria to grow on minimal medium, the bacteria need to encode for all 5 essential amino acids.



Which process or processes could explain these results?

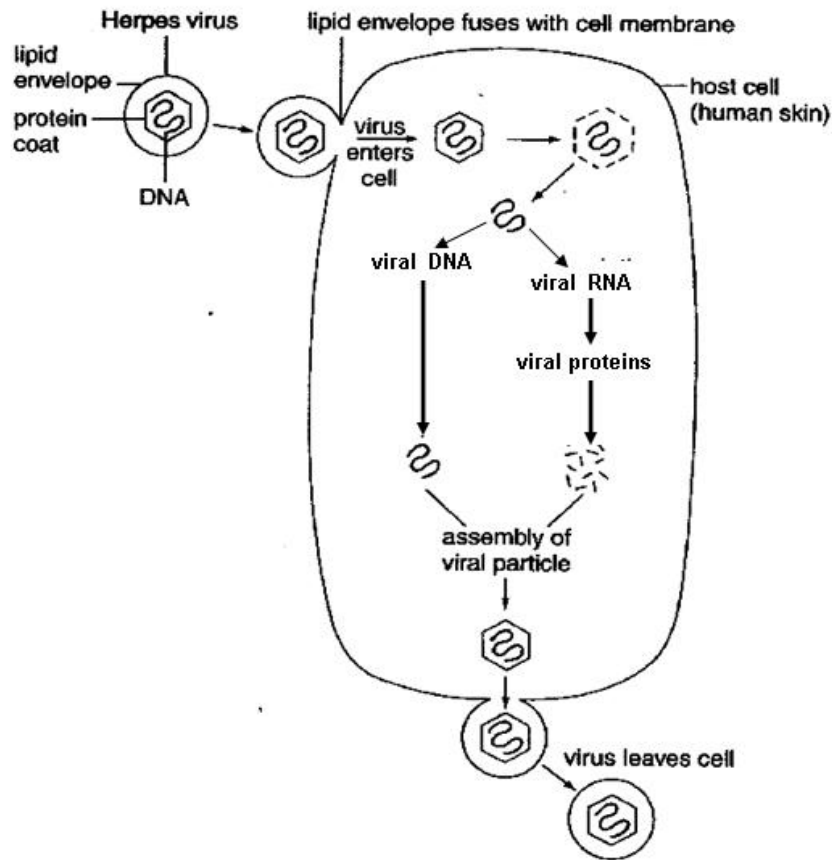
- 1 Conjugation
- 2 Transduction
- 3 Transformation

- A 1 only
 B 3 only
 C 1 and 2
 D 1 and 3

[Turn Over

- 13** A single mutation has occurred in an *E. coli* cell. If the structural genes in a *trp* operon were not expressed regardless of the presence or absence of the amino acid tryptophan, which of the following mutations could have taken place?
- 1 The cell had a mutation at the operator of the *trp* operon.
 - 2 The cell had a mutation at the *trpR* gene.
 - 3 The cell had a mutation at the promoter of the *trp* structural genes.
- A** 3 only
B 1 and 2
C 2 and 3
D 1, 2 and 3
- 14** When the *lac* operon for lactose metabolism is switched off, which of the following genes would still be expressed?
- I β -galactosidase gene
 - II RNA polymerase gene
 - III CAP gene
 - IV Repressor gene
- A** I and II
B I and III
C II, III and IV
D All of the above
- 15** Which features of viruses account for them being obligate parasites?
- 1 All viruses are very small, ranging in size from 20 – 300nm.
 - 2 Each virus contains only one type of nucleic acid.
 - 3 Viruses can be crystallised.
 - 4 Viruses cannot synthesis ATP.
 - 5 Viruses have no cellular structure.
 - 6 Viruses have no enzymes involved in metabolism outside a host cell.
- A** 1, 3 and 5
B 1, 4 and 6
C 2, 3 and 5
D 2, 4 and 6

- 16 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth. With reference to the diagram below, which of the following statements best describes the herpes virus?



- A It is not a retrovirus as it does not contain RNA as its genetic material.
- B Its mode of replication is similar to that of HIV.
- C Its replication cycle includes a lysogenic phase.
- D Death of the host cell is necessary for the release of the viral progeny.

[Turn Over

- 17** The MDR1 gene codes for a membrane transport protein in mammalian cells. The protein has the ability to prevent the entry of, or remove, molecules such as chemotherapeutic drugs, from cells. Gene amplification of the MDR1 gene can arise from environmental signals.

Which statement is not consistent with the information above?

- A** Amplification of the MDR1 gene in mammalian cancer cells could lead to the cells becoming resistant to chemotherapeutic drugs.
- B** Cancer cells, with initially the same level of MDR1 gene expression as healthy cells, may respond to chemotherapy by increasing the number of MDR1 genes, which increases expression.
- C** Development of chemotherapeutic drug resistance can be progressive, rather than spontaneous, and can hinder chemotherapy in patients with cancer.
- D** Since gene amplification has not arisen by genetic causes, descendant cells from a cancer cell with MDR1 gene amplification will be normal for MDR1 gene expression.

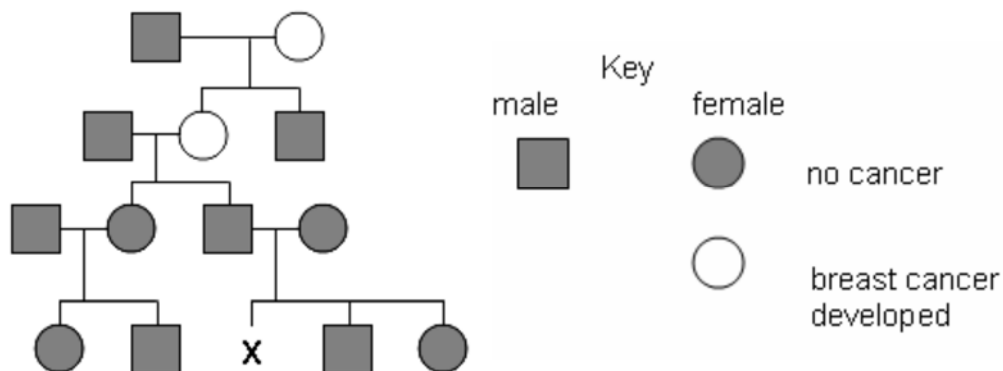
- 18** Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were unable to produce the protein, RB.

Addition of RB to these cells reduced their rate of division.

Which of the following statement accounts for the observation?

- A** Both chromosomes in the cancer cell carry alleles for tumour suppressor gene.
- B** Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.
- C** Both chromosomes in the cancer cell carry alleles for proto-oncogene.
- D** Both chromosomes in the cancer cell have the allele for proto-oncogene deleted.

- 19 The diagram shows the inheritance of a form of breast cancer associated with the presence of just one allele of the autosomal gene BRCA 1.



What is the probability that woman X inherits the BRCA 1 allele associated with breast cancer?

- A 0.00
 B 0.25
 C 0.50
 D 1.00
- 20 The table summarises a breeding experiment with *Drosophila melanogaster*.

parents	BBNN x bbnn			
F1	BbNn			
test cross	BbNn x bbnn			
test cross offspring	BbNn	bbNn	Bbnn	bbnn
number of test cross offspring	493	125	101	585

key

- B wild type body colour N normal wings
 b black body colour n vestigial wings

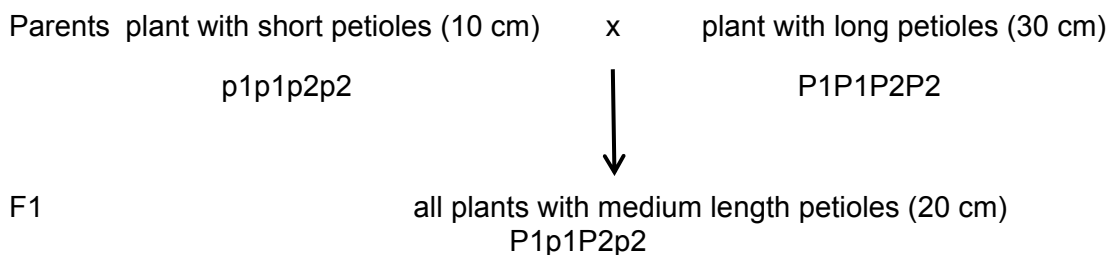
What do these results suggest?

- A The alleles N and n are linked.
 B Independent assortment has not occurred.
 C The alleles for normal and vestigial wings are codominant.
 D The gene loci for body colour and wing shape are not linked.

[Turn Over

- 21** The length of the petiole (leaf stalk) in a type of flowering plant is controlled by two genes, P1 and P2. These genes are at different loci on non-homologous chromosomes. Plants with long petioles (30 cm) are homozygous dominant. Plants with short petioles (10 cm) are homozygous recessive. Each dominant allele contributes an equal length to the petiole.

To obtain plants which have medium length petioles (20 cm), a plant breeder carries out the cross below.



If the F1 generation plants with medium length petioles were allowed to cross, what proportion of their offspring would be expected to have medium length (20 cm) petioles?

- A 0.0625
 B 0.250
 C 0.375
 D 0.5
- 22** The diagram shows the phenotypes of the different shape combs that some breeds of chicken have on their heads.



A cross between pure-breeding chickens with rose combs and pure-breeding chickens with pea combs, gave F1 generation offspring with walnut combs.

Interbreeding these F1 generation chickens with walnut combs produced offspring which showed all four of the possible comb phenotypes.

What is the expected phenotypic ratio of the offspring of a cross between a chicken with a single comb and one of the F1 generation chickens with a walnut comb?

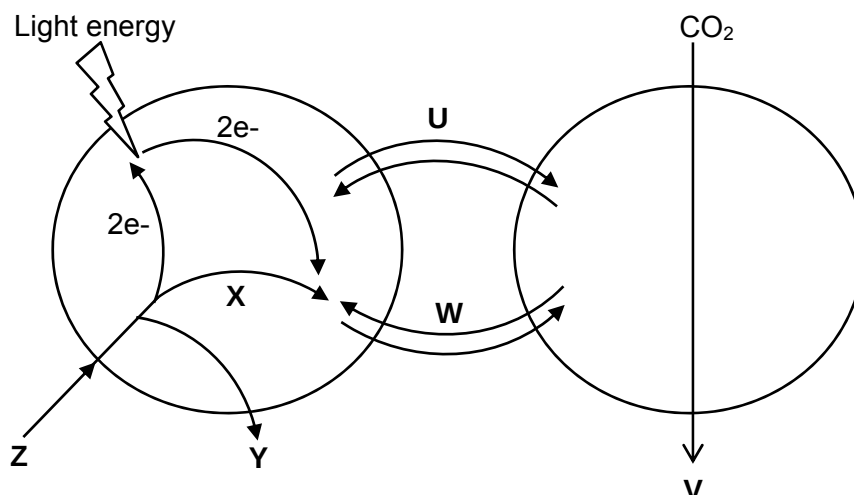
- A 1 rose : 1 pea : 1 walnut : 1 single
 B 1 rose : 2 walnut : 1 pea
 C 1 walnut : 1 single
 D 9 walnut : 7 single

- 23** The enzyme phosphofructokinase is involved in the phosphorylation of hexose phosphate sugars during glycolysis. It is involved in the control of the rate of glycolysis, and thus respiration, by end-product inhibition.

Which of the following is a description of this enzyme?

	shape of binding site(s)	substrate	products
A	no allosteric site, active site complementary to ATP and hexose	hexose	hexose phosphate
B	allosteric site complementary to glucose active site complementary to hexose phosphate	hexose phosphate	hexose
C	allosteric site complementary to ATP, active site complementary to ATP and hexose phosphate	hexose phosphate	hexose bisphosphate
D	no allosteric site, active site complementary to hexose bisphosphate	hexose bisphosphate	fructose bisphosphate

- 24** The diagram summarises the reactions of photosynthesis in a plant.

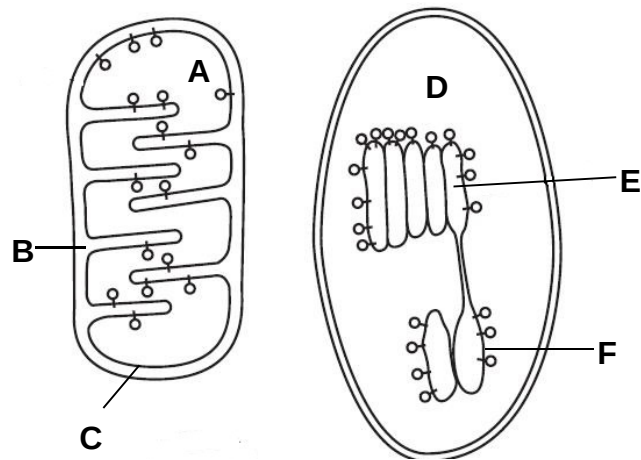


Which of the following correctly identifies the substances involved?

	U	V	W	X	Y	Z
A	triose phosphate	cellulose	ATP	H^+	O_2	H_2O
B	NADP+	triose phosphate	ATP	H^+	H_2O	O_2
C	NADPH	triose phosphate	ADP	H^+	O_2	H_2O
D	NADPH	cellulose	ADP	H^+	H_2O	O_2

[Turn Over

- 25 Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another. Figure below shows two such membrane systems.



Which statements about the two membrane systems are correct?

- 1 There is production of NADH and ATP within structure F.
- 2 Structure D is the location in which glyceraldehyde-3-phosphate is produced.
- 3 Protons are pumped from structure D to structure E against its electrochemical gradient via ATP synthase on structure F.
- 4 Electrons flow from structure B to structure A down its electrochemical gradient via cytochromes.
- 5 Protons diffuse down its proton gradient via ATP synthase on structure C and ATP are produced in structure A.

- A 1 and 4
 B 2 and 5
 C 1, 2 and 5
 D 2, 3 and 5

26 The table below shows a description of the activity of 4 drugs.

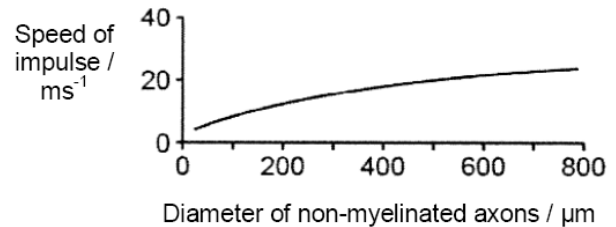
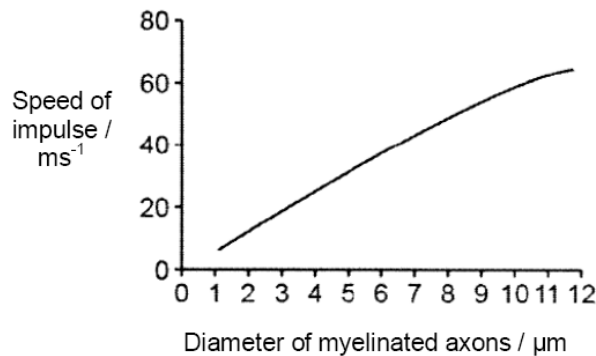
Drug	Description
1	Inhibit cAMP synthesis
2	Inhibit acetylcholinesterase
3	Block ligand gated Ca^{2+} channels
4	Block K^+ channels

Which of the following combination shows the consequence for each of the four drugs?

	Drug 1	Drug 2	Drug 3	Drug 4
A	Decreased action potential initiation	Decreased signal transduction efficiency	Decreased action potential initiation	Increased action potential initiation
B	Decreased activation of signalling pathways	Increased action potential initiation	Decreased action potential initiation	Delayed response to stimulus
C	Decreased activation of signalling pathways	Decreased action potential initiation	Increased action potential initiation	Increased action potential initiation
D	Decreased signal transduction efficiency	Increased action potential initiation	Increased action potential initiation	Delayed response to stimulus

[Turn Over

- 27** The ability of organisms to respond rapidly to stimuli is limited by the speed of the impulses in their neurons. The axons of invertebrate neurones lack a myelin sheath. The axons of most vertebrate neurons are myelinated. The graphs show the speed of impulses in these two types of axon.

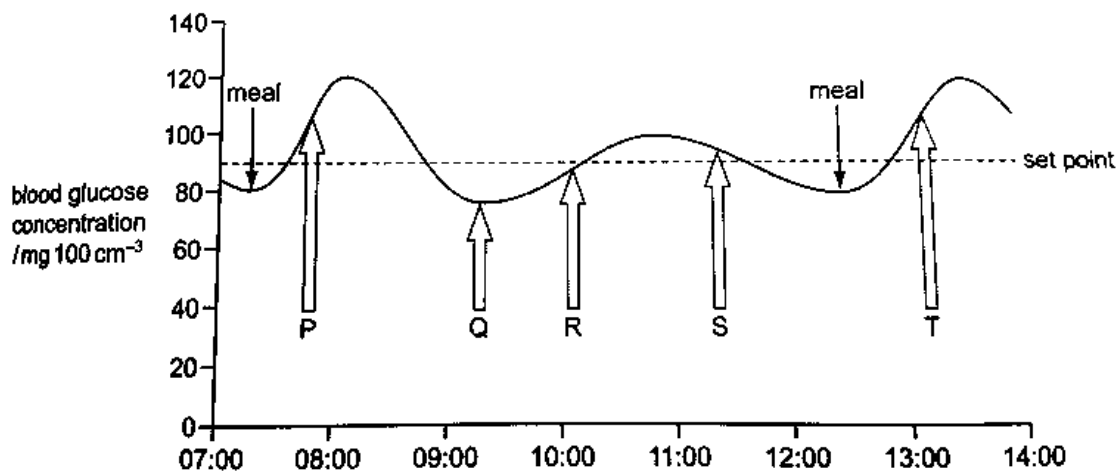


Which of the following statements are correct?

- 1 The action potential in myelinated axons is greater than the action potential in non-myelinated axons.
- 2 The speed of impulses is changed by the diameter of the axon.
- 3 Increasing the diameter of a myelinated axon causes a greater change in the speed of conduction than increasing the diameter of a non-myelinated axon.
- 4 The presence of myelin increases the speed at which impulses are conducted.

- A** 1, 2 and 3 only
B 1, 2 and 4 only
C 2, 3 and 4 only
D 3 and 4 only

- 28 The graph shows the way in which blood glucose concentrations vary over part of a day for a person who does not have diabetes. The set point, indicated by a dotted line, is the blood glucose concentration that is maintained by homeostasis. During this time the person has two meals.



Which row correctly identifies times at which insulin secretion, glucose absorption from the small intestine and conversion of glucose to glycogen occur?

	Insulin secretion	Glucose absorption	Glucose converted to glycogen
A	P, S and T	P and T	P, S and T
B	P, R and T	P and T	S
C	P, S and T	P, R and T	Q and S
D	R and S	P, Q, R and T	P, Q, S and T

- 29 Which statements are acceptable parts of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species perfectly adapted to a stable environment will continue to evolve.
- 4 Variation between individuals of a species is essential for evolutionary change.

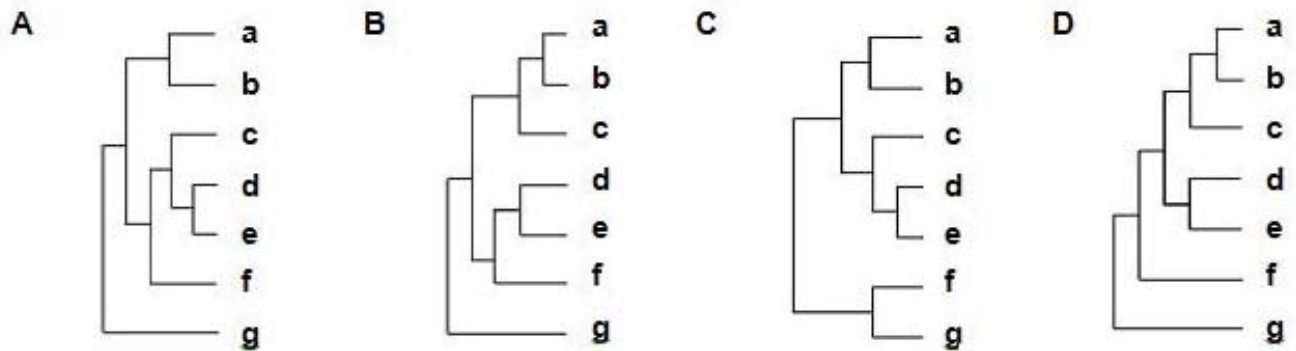
- A** 1, 2 and 4 only
B 2 and 3 only
C 3 and 4 only
D 4 only

[Turn Over]

- 30 The table shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

	b	c	d	e	f	g
a	39	72	128	126	127	269
b		81	130	128	129	268
c			129	127	128	267
d				56	154	271
e					151	268
f						273

Which of the following phylogenetic trees best shows the relationship among these seven species?

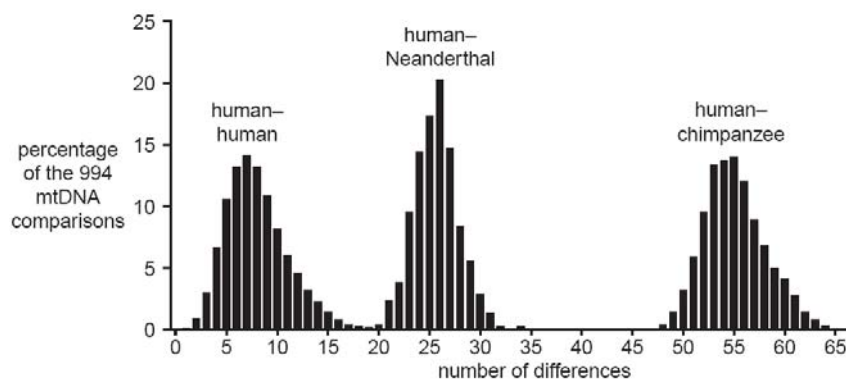


31 A scientist isolated and analysed the nucleotide sequence of the mtDNA of

- 994 modern humans
- One Neanderthal fossil
- Nine chimpanzees.

They compared these sequences. They were able to make human to human, Neanderthal to human and chimpanzee to human comparisons. The number of differences in the nucleotide sequence for each comparison was then recorded. The differences in mtDNA were quite small – no more than a few nucleotide bases – and relatively neutral in terms of evolution.

The results of these comparisons are shown in the graph below.

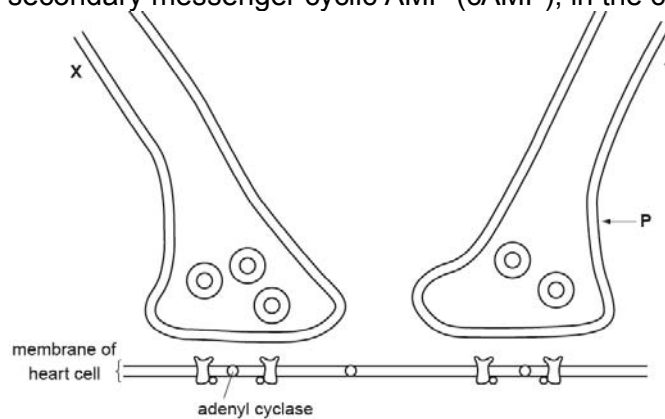


Which of the following statements is **not** true?

- A** This is an example of continuous variation.
- B** The difference between human mtDNA and that of the chimpanzees was greater than between Neanderthals and humans.
- C** Neanderthals are evolutionarily closer to humans than to chimpanzees.
- D** The differences between the DNA of modern humans are much less than the differences between modern humans and Neanderthals.

[Turn Over

- 32 Figure below shows ending of neurone Y in the heart. Stimulation of Y leads to an increase in the concentration of the secondary messenger cyclic AMP (cAMP), in the cytoplasm of the heart cell.



The stages involved in this cell signaling process include:

- 1 Vesicles fuse with post-synaptic membrane and exit via exocytosis and diffuse across the synaptic cleft.
- 2 Voltage-gated calcium channel opens leading to an influx of calcium ions into the neurone.
- 3 Adenyl cyclase becomes activated and converts ATP to cAMP.
- 4 Calcium ions cause movement of neurotransmitter vesicles towards post-synaptic membrane (along microtubules).
- 5 Neurotransmitter binds to G protein coupled receptor on the membrane of heart cell and activates G protein.

What is the sequence of these stages?

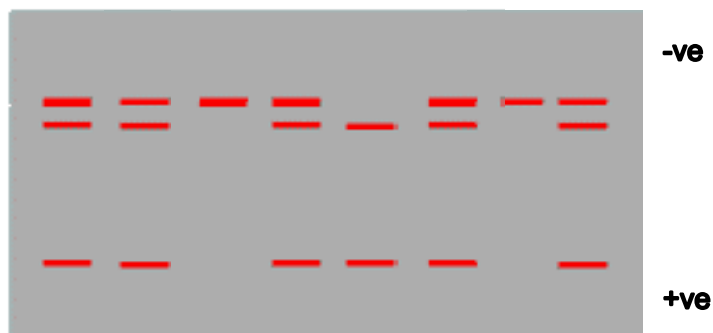
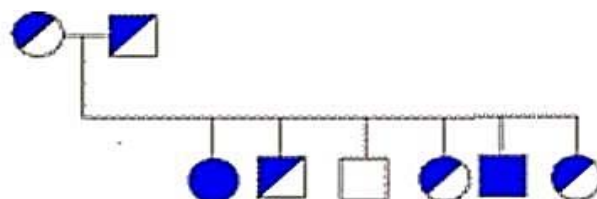
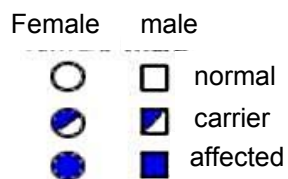
- A 5 → 2 → 4 → 1 → 3
- B 5 → 2 → 4 → 3 → 1
- C 2 → 4 → 1 → 5 → 3
- D 2 → 1 → 4 → 5 → 3

- 33 Chemical T is known to act on the cell signaling pathway of glucagon. It is found to diminish the effect of glucagon.

Which of the following is **not** a possible effect of Chemical T?

- A Chemical T interferes with release of glucagon from the pancreas.
- B Chemical T activates the hydrolysis of GTP on the G protein.
- C Chemical T inactivates the enzyme adenylyl cyclase.
- D Chemical T act as a competitive inhibitor of the protein kinase.

- 34 The diagram below shows an autoradiograph, post-gel electrophoresis, showing the restriction fragments obtained from various members of a family with respect to a disease.



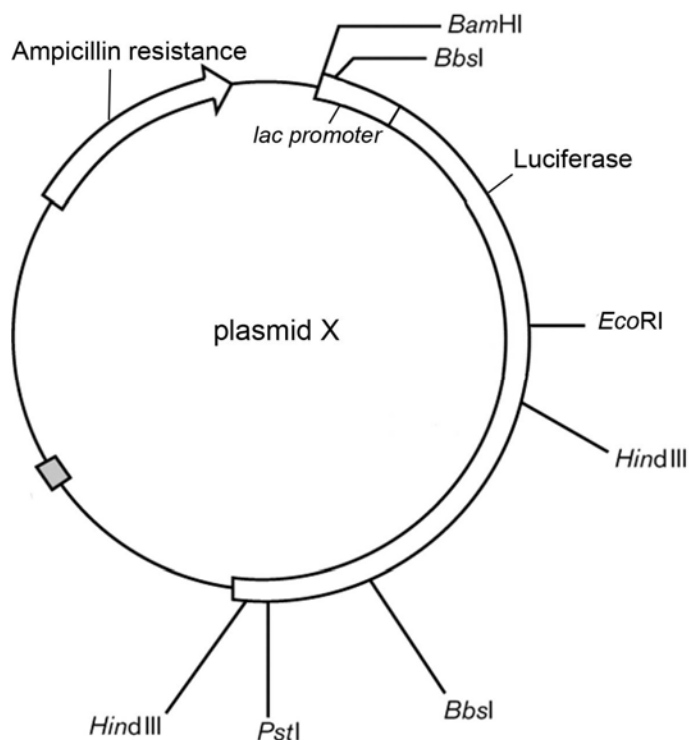
Which of the following statements can be concluded from this autoradiograph?

- 1 The recessive allele is missing a restriction site.
- 2 The disease is autosomal recessive.
- 3 The radioactive probe is able to bind to a specific sequence on the recessive allele only.

- A** 1 and 2
B 2 and 3
C 1 and 3
D All of the above

[Turn Over

35 Plasmid X can serve as a vector for the insertion of genes to be cloned.



Which of the following options will allow the selection of the colonies containing the recombinant form of plasmid X?

	Selection medium	Phenotype of colonies that contain the inserted gene
A	Containing ampicillin and lactose	White colonies
B	Containing ampicillin and luciferase	Colonies that emit light
C	Containing ampicillin, lactose and luciferin	White colonies
D	Containing ampicillin, lactose and luciferin	Colonies that emit light

- 36 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'

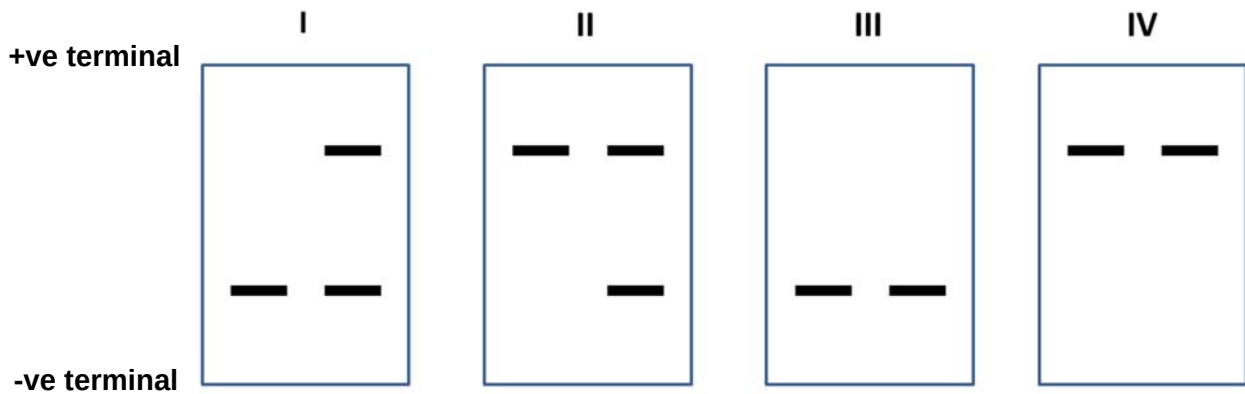
3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A 3' TAAGCCT 5' & 5' CTAGAGG 3'
- B 5' ATTCGGA 3' & 3' GATCTCC 5'
- C 3' UAAGCCU 5' & 5' CUAGAGG 3'
- D 5' AUUCGGA 3' & 3' GAUCUCC 5'
- 37 Cystic fibrosis (CF) is an autosomal recessive genetic disorder. An individual must have two copies of the mutated CFTR gene to express the disease phenotype. One of the most common CF-causing mutation resulted in a loss of phenylalanine located at position 508 of the protein.

The DNA sequence of the CF locus from the offspring of 2 carriers are removed and separated by gel electrophoresis.

Which pattern of bands corresponds to two of the offspring that are phenotypically normal?



- A I only
- B II only
- C I and III
- D II and IV

[Turn Over

38 The statements are about the preparation and application of DNA libraries.

- 1 A cDNA library allows the study of the functions of introns of specific genes
- 2 A genomic library enables detection of genes that, in the host, have no detectable level of expression.
- 3 Alternative splicing can be studied using a cDNA library
- 4 The preparation of a genomic DNA library requires restriction enzyme, reverse transcriptase and DNA ligase

Which statements are correct?

- A** 1, 2 and 4
- B** 1 and 4 only
- C** 2, 3 and 4
- D** 2 and 3 only

39 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell.

Which statement explains this?

- A** Different stem cells have only the genes required for their particular cell line.
- B** Genes not required for a particular cell line are methylated.
- C** Genes not required for a particular cell line are removed using restriction enzymes.
- D** mRNA that is not required for a particular cell line is destroyed.

40 Gene therapy is used to treat inherited diseases such as cystic fibrosis. Some of the scientific and ethical concerns about gene therapy are listed below.

- 1 Most gene therapy must be repeated in succeeding generations since germ cells are not involved.
- 2 Genetically modified organisms used in producing the gene therapy may escape into the environment with unforeseen consequences.
- 3 Putting genes into the germ line affects subsequent generations and is banned in many countries.
- 4 The same techniques for treating serious, life-threatening conditions may be used to try to change other things such as intelligence and skin colour.
- 5 Viral vectors, such as those used in the treatment of cystic fibrosis, have been known to produce harmful side-effects.

Which row identifies the types of concern?

	Scientific concerns	ethical concerns
A	1 and 2	3, 4 and 5
B	2 and 4	1, 3 and 5
C	1 and 5	2, 3 and 4
D	3, 4 and 5	1 and 2

BLANK PAGE